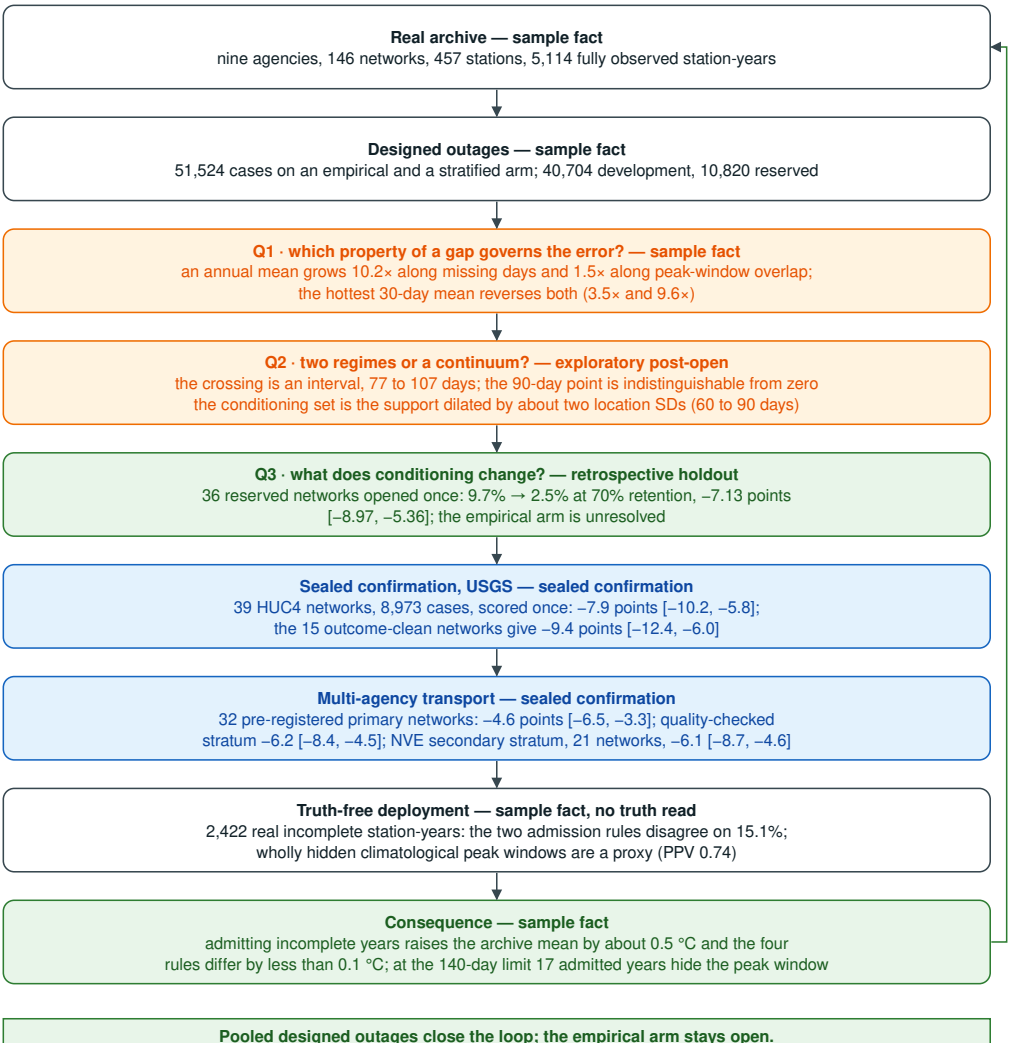
Real archive — sample fact
nine agencies, 146 networks, 457 stations, 5,114 fully observed station-years
Designed outages — sample fact
51,524 cases on an empirical and a stratified arm; 40,704 development, 10,820 reserved
Q1 · which property of a gap governs the error? — sample fact
an annual mean grows 10.2× along missing days and 1.5× along peak-window overlap;
the hottest 30-day mean reverses both (3.5× and 9.6×)
Q2 · two regimes or a continuum? — exploratory post-open
the crossing is an interval, 77 to 107 days; the 90-day point is indistinguishable from zero
the conditioning set is the support dilated by about two location SDs (60 to 90 days)
Q3 · what does conditioning change? — retrospective holdout
36 reserved networks opened once: 9.7% → 2.5% at 70% retention, –7.13 points
[–8.97, –5.36]; the empirical arm is unresolved
Sealed confirmation, USGS — sealed confirmation
39 HUC4 networks, 8,973 cases, scored once: –7.9 points [–10.2, –5.8];
the 15 outcome-clean networks give –9.4 points [–12.4, –6.0]
Multi-agency transport — sealed confirmation
32 pre-registered primary networks: –4.6 points [–6.5, –3.3]; quality-checked
stratum –6.2 [–8.4, –4.5]; NVE secondary stratum, 21 networks, –6.1 [–8.7, –4.6]
Truth-free deployment — sample fact, no truth read
2,422 real incomplete station-years: the two admission rules disagree on 15.1%;
wholly hidden climatological peak windows are a proxy (PPV 0.74)
Consequence — sample fact
admitting incomplete years raises the archive mean by about 0.5 °C and the four
rules differ by less than 0.1 °C; at the 140-day limit 17 admitted years hide the peak window
Pooled designed outages close the loop; the empirical arm stays open.